

R4: Graduation project

UK Railway Project

Presentation

Presented By:

Habiba Waleed

Team Members



Habiba Waleed

**Habiba
Mohammed**

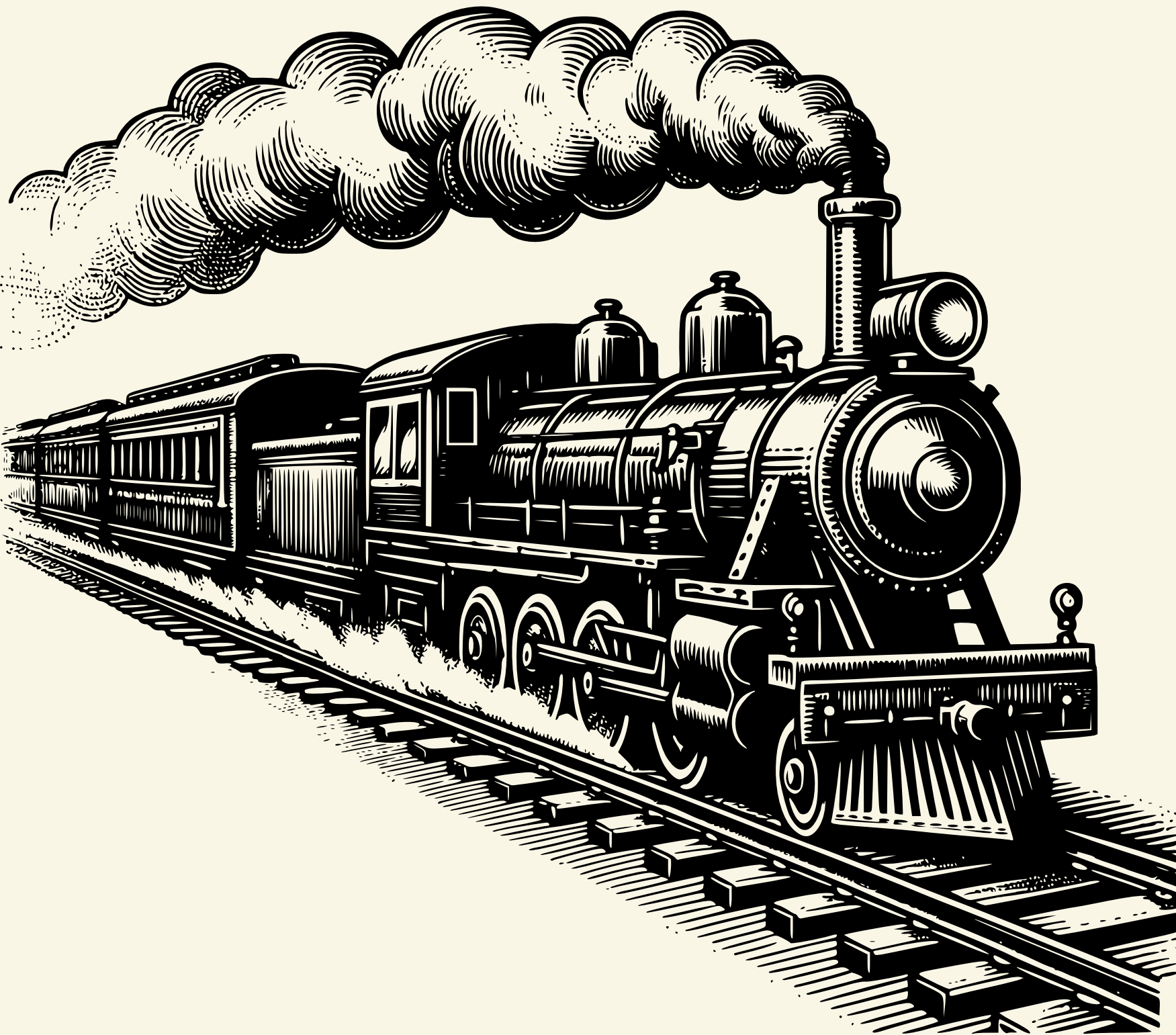
**Rodayna
Mahmoud**

Aya Hossam

**Sara
Mohammed**

Ganna Salah

Roles of Team Members



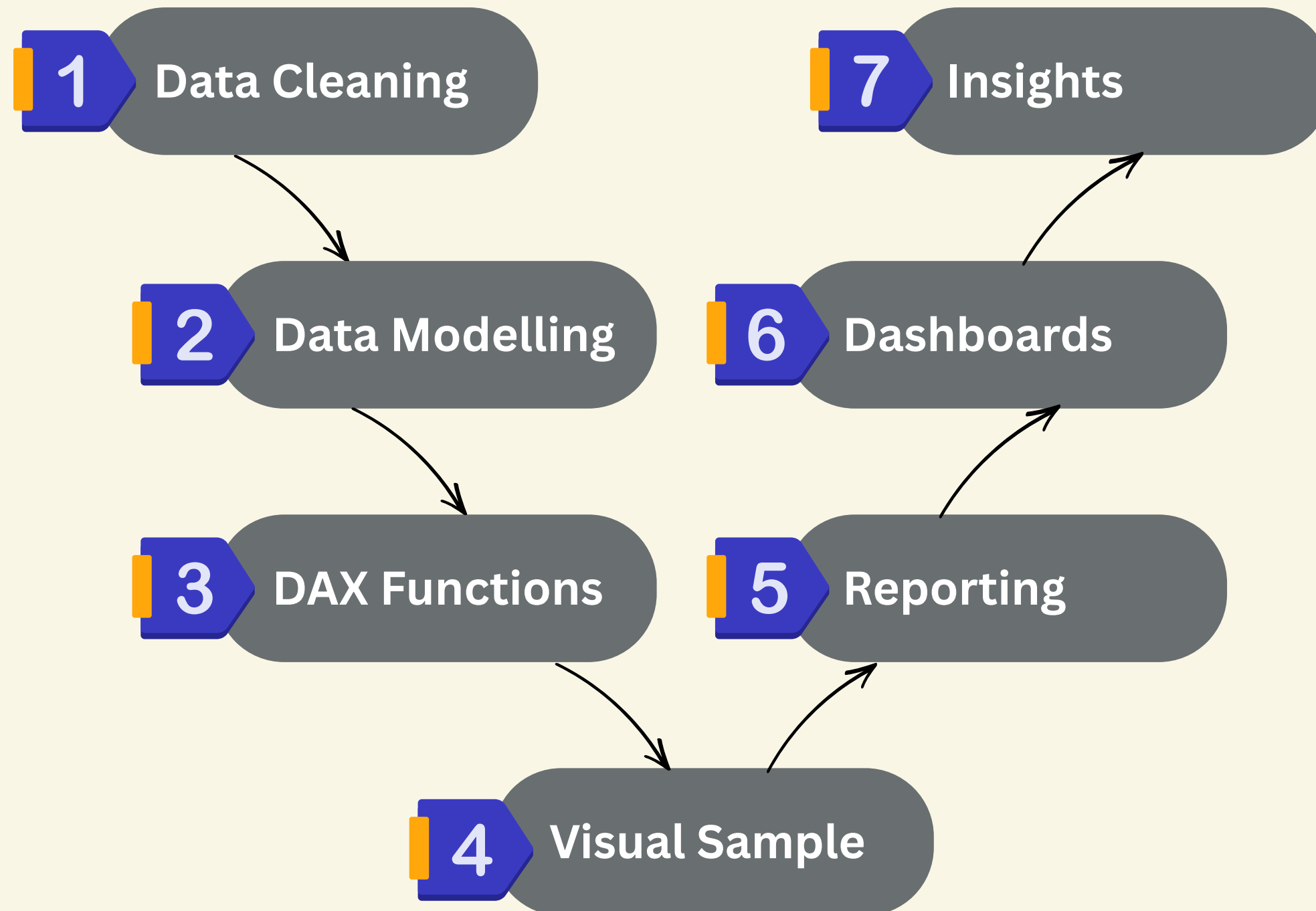
Team Members	
Name	Role
Habiba Waleed Abdelrazek Mohammed	Project Leader, Forecasting Dashboard
Habiba Mohammed Abdelfattah	Dashboard Visualization
Gannah Salah	Refund& financial analysis Dashboard
Aya Hossam Mostafa	Analysis and insights
Rodayna	Data cleaning and preparation
Sara Mohamed	Cancelled Rides
Each one take part of it depend on the dashboard page	Documentation & Presentation



Dashboard Pages

#	Page	What it covers
1	Financial Performance	Total revenue · MoM trends · Ticket type & payment method breakdown
2	Tickets Details	Ticket class & type demand · Time-of-day segments · Decomposition Tree
3	Delayed Rides	Delay counts · Root causes · Station breakdown · Ticket class analysis
4	Cancelled Rides	Cancellation counts · Reasons · Route & ticket type impact
5	Refunds & Financial Analysis	Refund rate · Revenue lost · Station-level refund breakdown
6	Forecasts	3-month projections for revenue, tickets, delays & cancellations

Analysis Steps



Project Overview

This Power BI project provides an in-depth analysis of UK train railway data, focusing on ticket purchases, journey details, and train performance. The dashboard enables users to explore trends in passenger travel, delays, ticket sales, and refund requests to improve operational insights.



Problem Statement

1 Revenue & Pricing

Which ticket types and routes drive the most revenue? How is revenue trending month-over-month?

2 Ridership & Demand

When and how do passengers travel?
Which stations are the busiest?

3 Operational Quality

How many journeys are delayed or cancelled, and why? What are the root causes?

4 Refunds & Loss

How much revenue is lost to refund requests?
Which disruptions drive the most claims?



Tools & Technologies Used



Power BI

Interactive visualizations and data modeling.



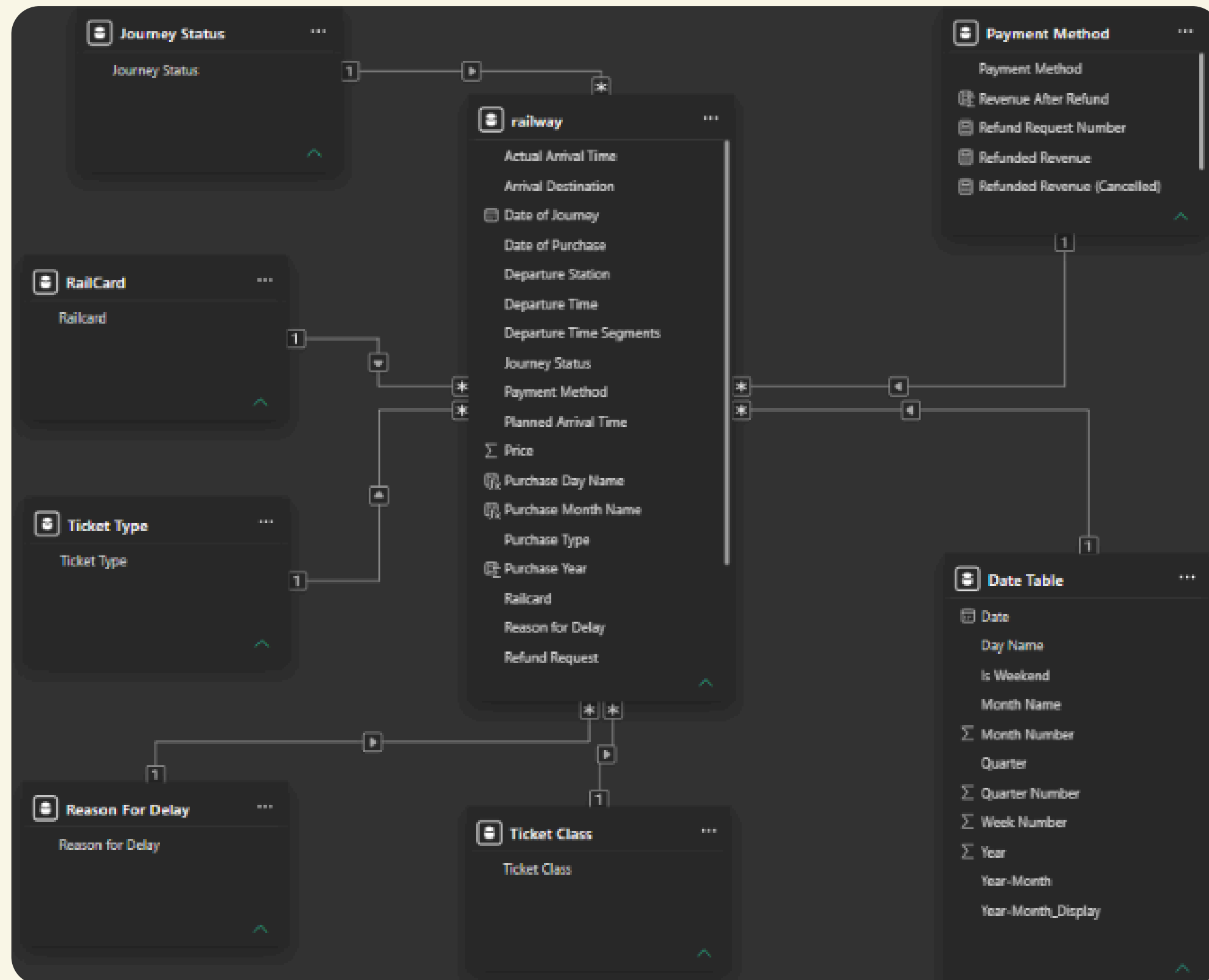
Power Query: Data transformation and cleaning.

$f(x)$

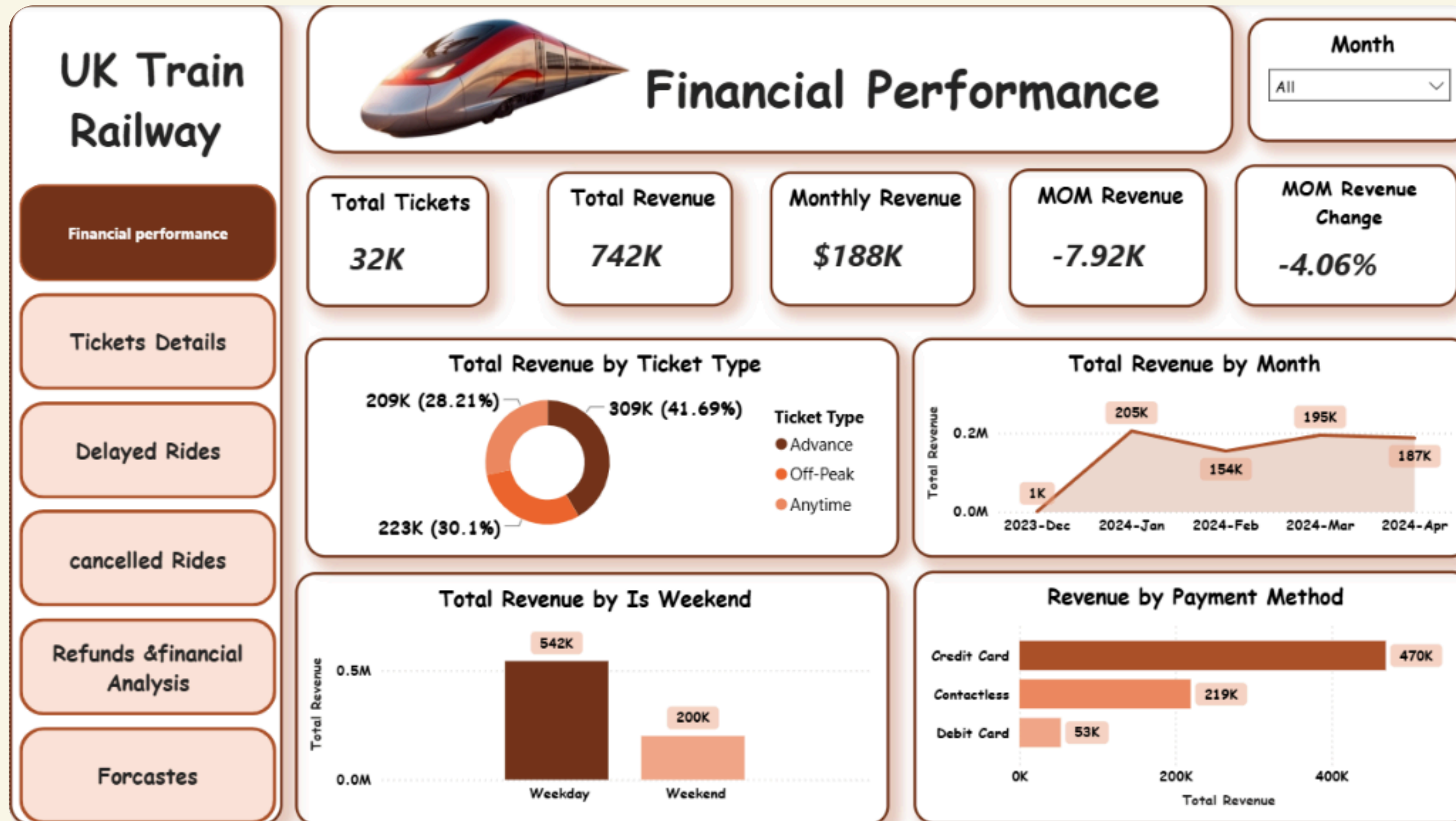
DAX (Data Analysis Expressions): Custom calculations and KPIs.



Star-Schema Modeling



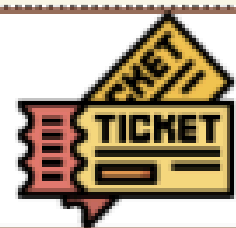
Financial Performance Dashboard



Financial Performance Insights

Insight	Tag	Solution
Total revenue reached \$742K across all months	Monitor	Establish a monthly revenue target dashboard to track performance against the \$742K baseline and identify underperforming months early.
Advance tickets generate the highest revenue at \$309K (41.69%)	Opportunity	Promote Advance tickets through loyalty programs and early-bird campaigns to maintain and grow this leading revenue stream.
Weekday trips significantly outperform weekends — \$542K vs \$200K	Action needed	Launch weekend-specific promotions, leisure bundles, and group travel offers to close the \$342K revenue gap.
Credit Card is the dominant payment method contributing \$470K	Diversify	Encourage contactless and digital wallet payments as faster alternatives to reduce over-reliance on a single payment method.
Revenue peaked in January 2024 at \$205K then declined in February before recovering	Investigate	Conduct root-cause analysis on the February dip — determine whether it is seasonal, pricing-related, or operational, and build a mitigation plan.
MoM Revenue Change was -4.06% indicating a slight downward trend in April	Action needed	Run targeted April promotions and monitor closely to distinguish a temporary dip from a sustained downward trend requiring structural changes.

Tickets Details Dashboard



Ticket Details

Total Tickets

32K

Avg Price

\$23.44

Payment Method

Contactless

Credit Card

Debit Card

Purchase Type

Online

Station

Ticket Type

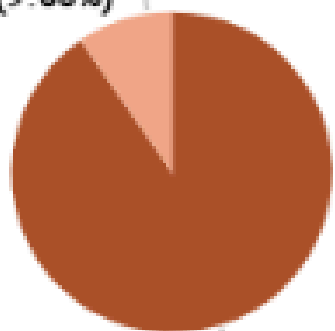
Advance

Anytime



Ticket Class Distribution

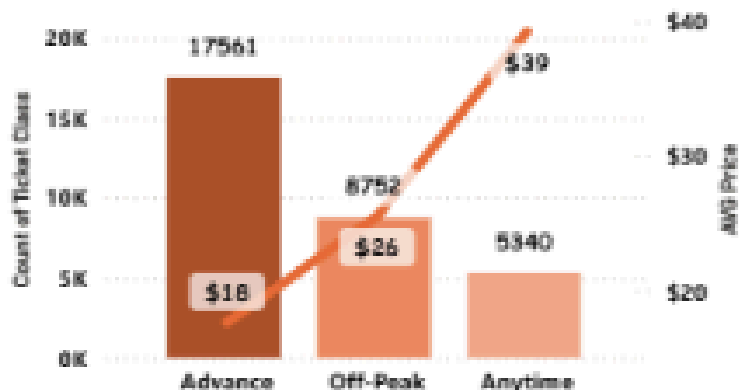
3.06K (9.66%)



Ticket Class
Standard
First Class

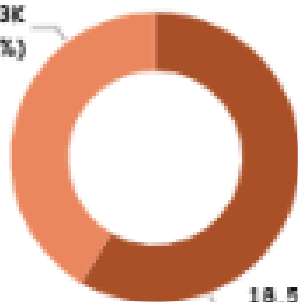
28.6K (90.33%)

Ticket Count & Average Price by Ticket Type



Ticket Sales by Purchase Type

13.13K (41.49%)



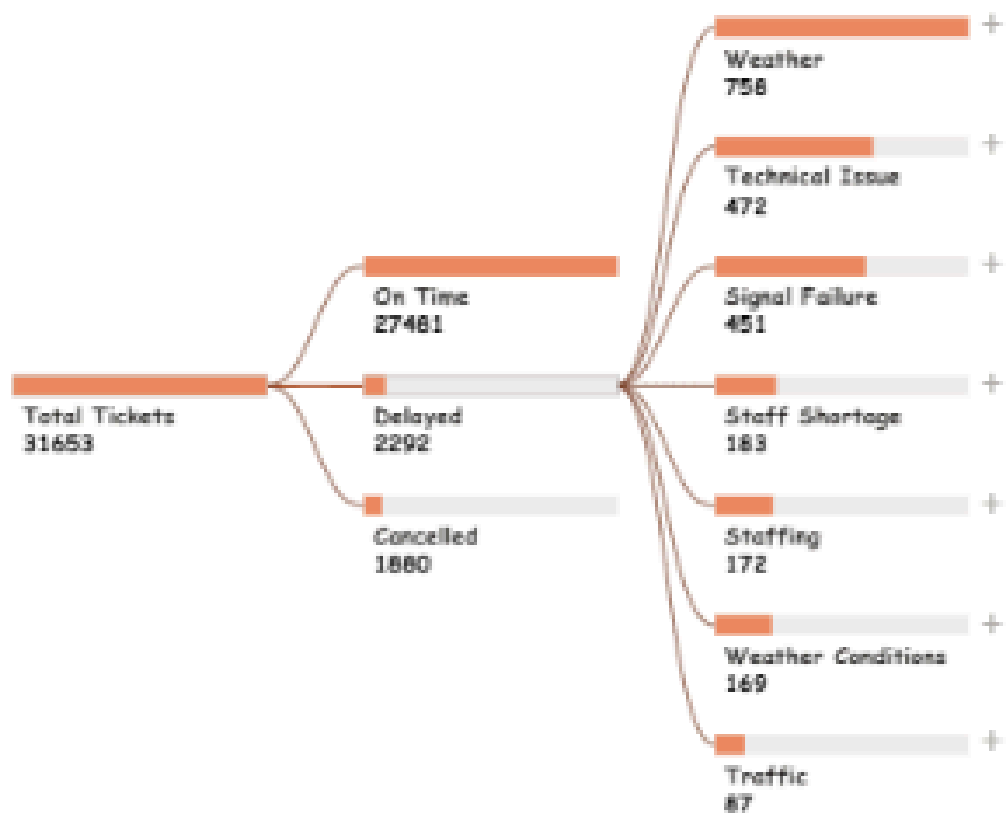
Purchase Type
Online
Station

18.52K (58.51%)

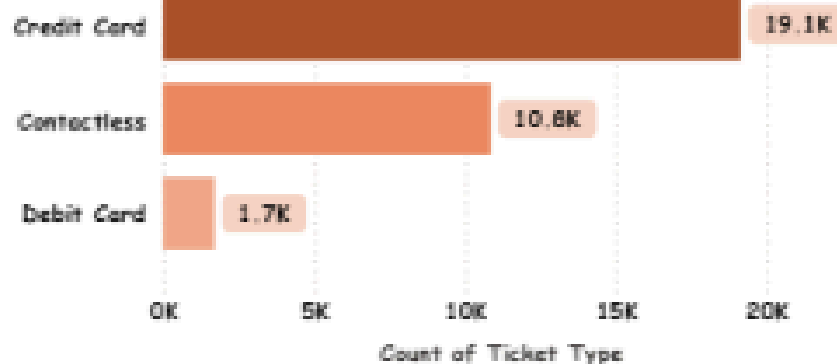
Journey Status

Reason for Delay

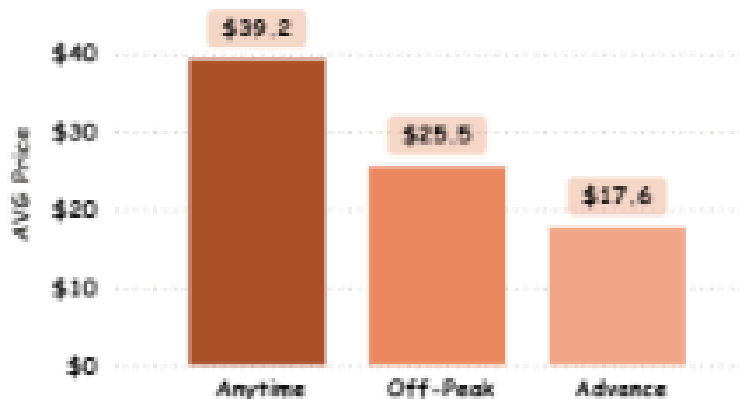
Delayed



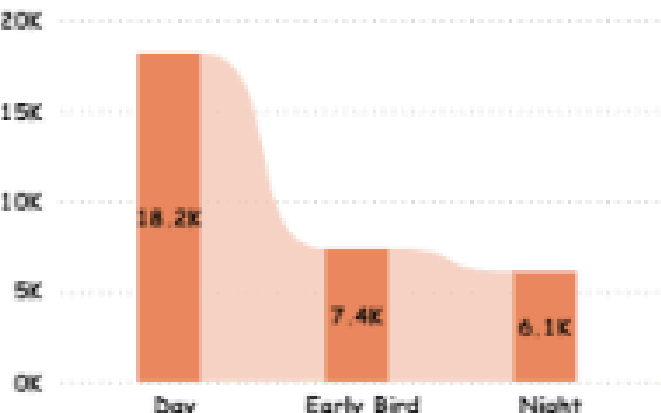
Ticket Count by Payment Method



AVG Price by Ticket Type



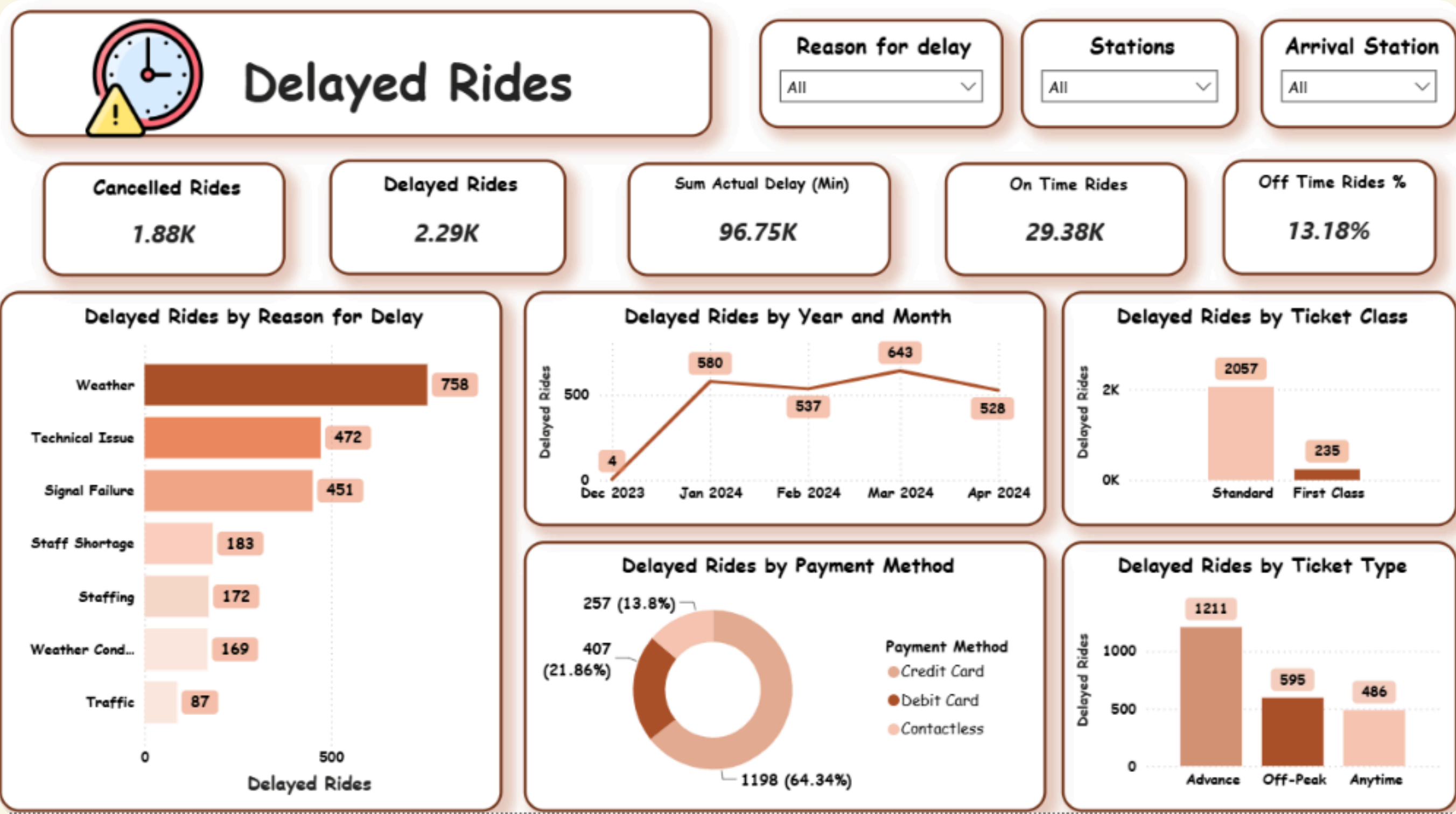
Tickets by Departure Time Segment



Tickets Details Insights

Insight	Tag	Solution
90.34% of passengers travel Standard Class vs only 9.66% First Class	Opportunity	Promote First Class upgrades during booking and check-in to increase revenue per ticket.
Advance tickets are the most purchased (17,561) but have the lowest average price (\$17.6)	Action needed	Introduce a small price increase by adding optional services such as seat selection or refreshments to increase the average order value while providing more value to customers.
Anytime tickets have the highest average price at \$39.2	Opportunity	Highlight their flexibility and convenience in marketing campaigns to justify the premium price and attract more business travellers.
Online purchases dominate at 58.51% vs 41.49% station purchases	Optimise	Continue improving the app and website to encourage more online bookings.
Daytime departures are the most popular with 18.2K tickets	Action needed	Add promotions for early morning and late evening departures to spread demand throughout the day.

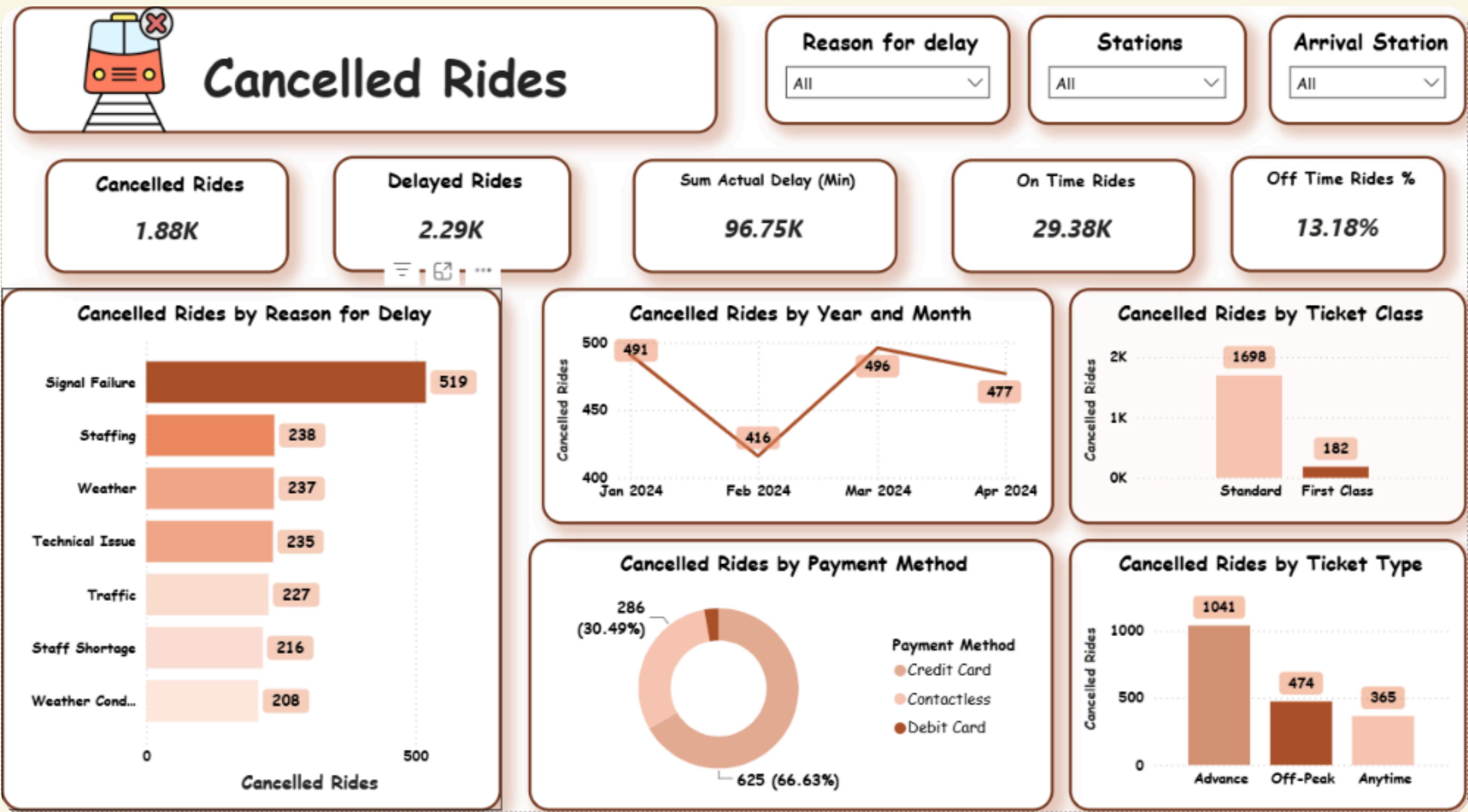
Delayed Rides Dashboard



Delayed Rides Insights

Insight	Tag	Solution
Total delayed rides reached 2,292 out of 31,653	Action needed	Set a formal KPI to track and reduce total delayed rides quarter-on-quarter; assign route-level accountability to operations managers.
Off-time rides percentage is 13.18%	Urgent	Target a reduction to below 10% within two quarters — any rate above 10% directly erodes passenger trust and drives refund costs.
Weather is the leading cause of delays with 758 rides affected	Action needed	Invest in real-time weather monitoring integrated with scheduling; trigger proactive passenger alerts before disruptions escalate.
Technical Issues rank second with 472 delayed rides	Action needed	Schedule regular preventive maintenance cycles and track MTBF per asset; reducing technical failures is the most controllable delay lever.
March 2024 recorded the highest delays at 643 rides	Investigate	Conduct a deep-dive on March operations and allocate additional staffing, fleet, and contingency resources for March in all future annual plans.
Standard Class passengers more affected — 2,057 vs 235 First Class	Review	Review whether First Class route prioritisation can be partially extended to Standard Class during high-delay periods.
Advance ticket holders experience the most delays (1,211 rides)	Action needed	Build a targeted real-time delay alert system for Advance holders; offer automatic rebooking options and compensation.
Total actual delay time reached 96.75K minutes	Monitor	Track average delay minutes per ride as a secondary KPI; set a quarterly reduction target for cumulative delay minutes.

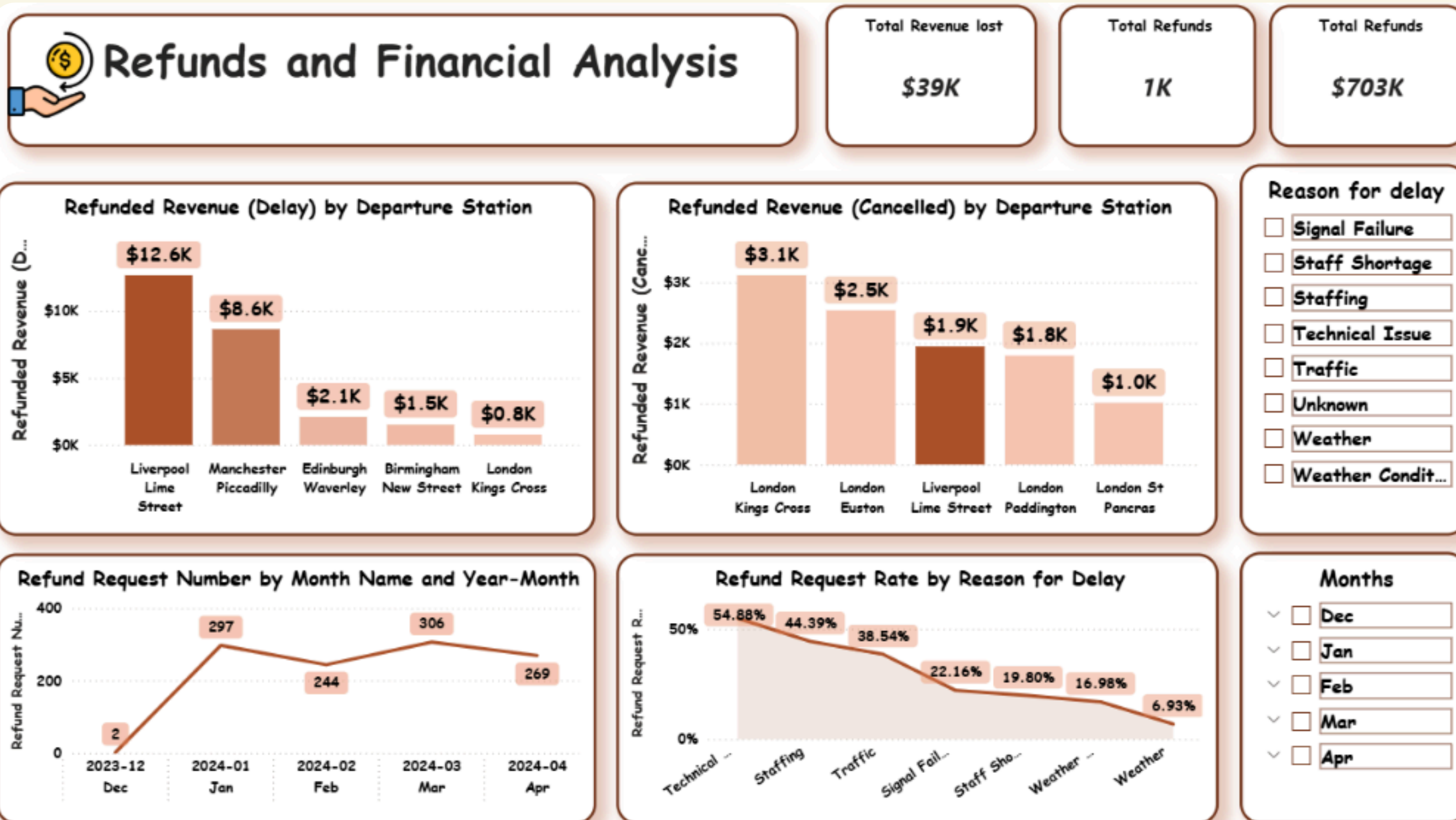
Cancelled Rides Dashboard



Cancelled Rides Insights

Insight	Tag	Solution
Total cancelled rides reached 1,880	Action needed	Benchmark against the industry cancellation average; set a quarterly reduction target with accountability per route and station.
Signal Failure is the primary cancellation reason with 519 rides	Urgent	Prioritise signal infrastructure upgrades — the single highest-impact intervention; negotiate reliability SLAs with network providers.
March 2024 had the highest cancellations at 496 rides	Investigate	Conduct root-cause analysis on the March spike; cross-reference with delay data to identify shared operational failures.
February 2024 had the lowest cancellations at 416 rides	Replicate	Analyse what operational conditions made February best-performing and use it as a blueprint for reducing cancellations in other months.
Standard Class passengers most affected with 1,698 cancellations	Action needed	Improve Standard Class cancellation handling — faster rebooking tools, clearer communication, and automatic compensation offers.
Advance ticket holders face the most cancellations (1,041 rides)	Action needed	Implement automatic rebooking and compensation vouchers for Advance holders on cancellation.
Credit Card users account for 66.63% of cancelled rides	Optimise	Ensure the refund process for card payments is fully automated, fast, and frictionless to protect customer trust.

Refunds & Financial Analysis



Refunds & Financial Analysis

Insight	Tag	Solution
Total revenue lost due to refunds reached \$39K	Action needed	Treat refund loss as a service quality KPI; link the \$39K figure directly to delay and cancellation reduction targets.
Total number of refund requests was 1,118	Automate	Set up automated refund processing triggered by confirmed delays and cancellations to reduce handling time and improve satisfaction.
Technical Issues trigger the highest refund request rate at 54.88%	Urgent	Fast-track technical issue resolution — more than half of all technical failures result in a refund; reducing failures is a direct financial recovery mechanism.
Staffing issues have the second highest refund rate at 44.39%	Action needed	Implement stricter staffing protocols and rostering oversight — staffing is the most controllable refund driver.
Liverpool Lime Street lost the most revenue from delay refunds at \$12.6K	Priority	Prioritise operational improvement at Liverpool Lime Street — the single largest refund recovery opportunity in the network.
Manchester Piccadilly ranks second in delay refunds at \$8.6K	Action needed	Run a combined improvement programme across Liverpool Lime Street and Manchester Piccadilly — they account for the bulk of delay refund losses.
London Kings Cross had the highest cancellation refunds at \$3.1K	Investigate	Analyse Kings Cross cancellation causes separately — the root cause likely differs and requires a tailored intervention.
Refund requests peaked in March 2024 at 306 requests	Priority	March is the highest month for delays, cancellations, and refunds — proactive operational mitigation in March is the highest-ROI intervention available.

Forecasting Dashboard



Forecasting Dashboard

Months

All

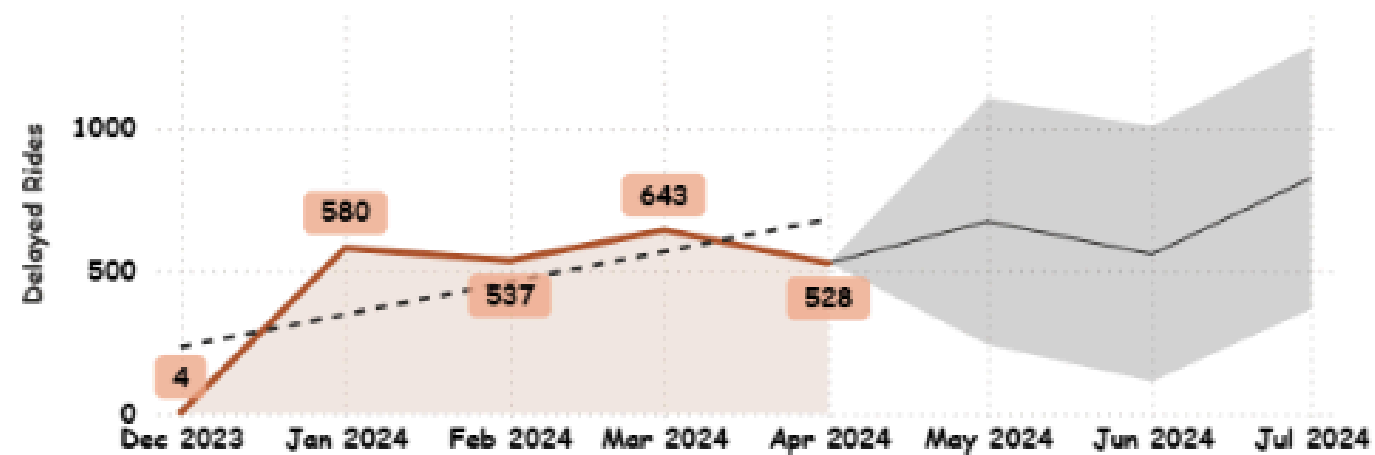
Day name

All

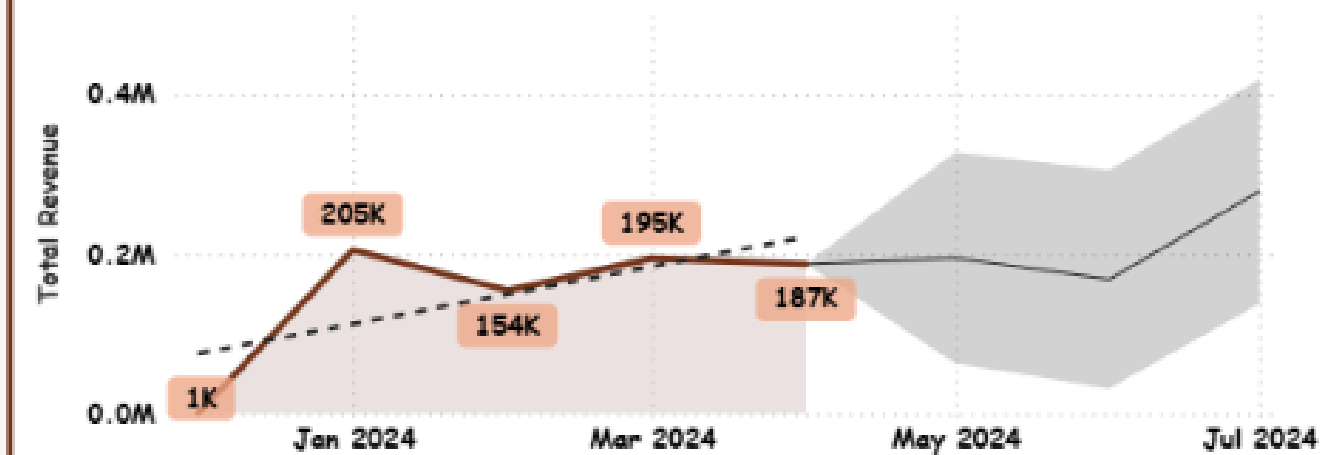
Weekend vs Weekday

All

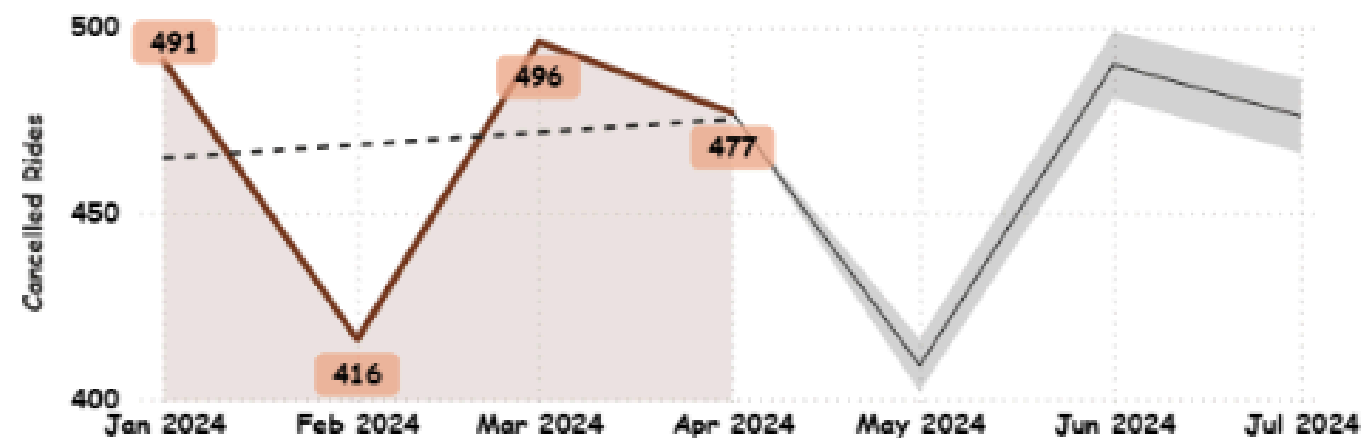
Next three months Delayed rides Forecast



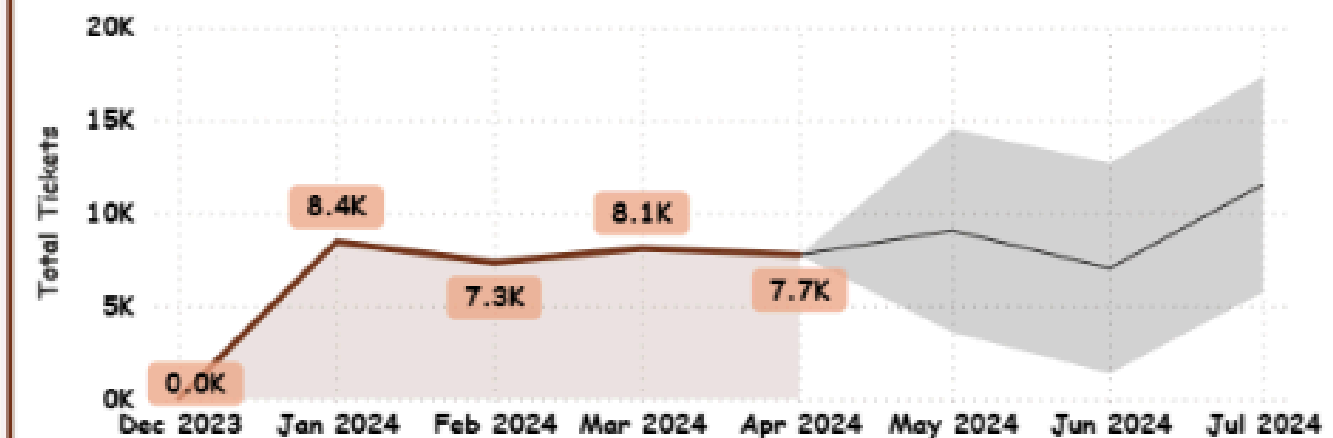
Next three months Revenue Forecast



Next three months Cancelled rides Forecast



Next Three Months Tickets Demand Forecast

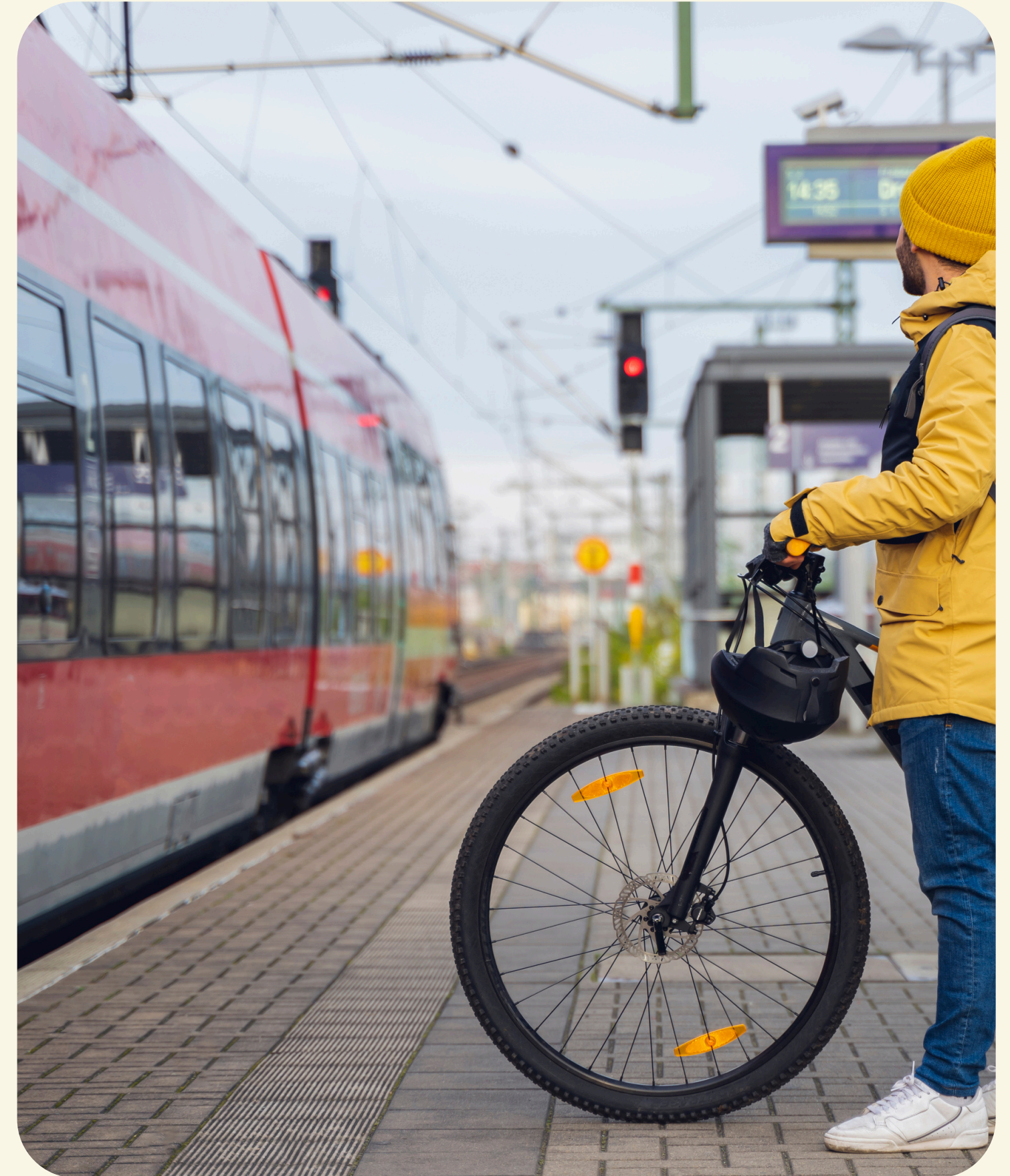


Forecasting Insights

Insight	Tag	Solution
Revenue is projected to increase over May–Jul 2024	Opportunity	Align marketing spend to these peak months; secure pricing agreements and capacity contracts ahead of demand to maximise revenue capture.
Ticket demand is expected to grow beyond the 7.7K–8.4K monthly range	Prepare	Scale ticketing infrastructure and train capacity ahead of the demand surge; avoid bottlenecks during high-volume periods.
Delayed rides forecast shows an upward trend continuing into mid-2024	Urgent	Act on delay root causes immediately — rising delays during a revenue growth period will compound refund costs and erode projected financial gains.
Cancelled rides are expected to rise significantly in June 2024	Urgent	Build June-specific contingency plans now — a surge in cancellations during a peak revenue month creates double impact: refund losses and reputation damage.

Key Features

- ➔ • **Ticket Sales Analysis:** Breakdown of ticket purchases by type, class, and payment method.
- ➔ • **Revenue Insights:** Total revenue tracking with month-over-month growth and payment channel breakdown.
- ➔ • **Journey Performance:** On-time performance, delay analysis, and cancellation patterns with root cause breakdown.
- ➔ • **Refund Analysis:** Financial impact of refund requests resulting from delays and cancellations, broken down by station and reason.
- ➔ • **Forecasting:** Next 3-month projections for revenue, ticket demand, delayed rides, and cancelled rides.



Key Insights



Revenue

Establish a monthly revenue target dashboard to track performance against the \$742K baseline and identify underperforming months early.



Operations

86.8% on time · 7.2% delayed · 5.9% cancelled ·
Weather = #1 root cause



Tickets

Advance tickets dominate (55% of sales, 42% of revenue). Standard class = 90% of journeys

- Promote Advance tickets through loyalty programs and early-bird campaigns to maintain and grow this leading revenue stream.



Refunds

3.5% refund rate · £38,702 refunded ·
Many disrupted passengers don't claim



Stations

Manchester Piccadilly busiest (5,650 journeys) ·
Top 5 hubs = 75% of all departures



Forecast

Revenue stable ~£185K–£200K/month ·
No deterioration in delay trend projected

Recommendations

1

Simplify refund claims

Many disrupted passengers don't file refunds — a simplified claims process could improve passenger trust and transparency.

2

Prioritise weather resilience

Weather is the #1 delay cause. Investment in weather-resilient infrastructure at the 5 major hubs would have the highest network-wide impact.

3

Promote First Class upselling

First Class is only 9.7% of journeys but commands higher prices. Targeted promotions during off-peak times could improve yield.

4

Grow weekend leisure travel

Weekend journeys represent 28.4% of volume. Discounted weekend Advance tickets could increase ridership without adding peak-hour pressure.

5

Address February revenue dip

February consistently underperforms. Targeted promotions or railcard incentives in January–February could soften the seasonal dip.



*Thank you
so much*

We seek better approaches towards
the future of sustainable, efficient rail.